

(1) Is the set of total functions from N to $\{T, F\}$ countable? Prove your answer.

The set of total function from N to $\{T, F\}$ is not countable.
 The proof is that total valued function correspond bijectly to subset N . The power set $|P(N)|$ has cardinality of $2^{\aleph_0}$ by Cantor's argument. Therefore set function is uncountable and this is same as more than $\aleph_0$ sets of integers.

Now, the set of all functions $f: N \rightarrow \{T, F\}$ is uncountable.

Let

$$S = \{f_i: N \rightarrow \{T, F\}\}.$$

Proof

Assume for contradiction that S is countable. Then we can list all such functions:

$$f_0, f_1, f_2, \dots$$

where each $f_i: N \rightarrow \{T, F\}$.

Define a new function $g: N \rightarrow \{T, F\}$ by

$$g(n) = \begin{cases} T & \text{if } f_n(n) = F, \\ F & \text{if } f_n(n) = T. \end{cases}$$

Equivalently, $g(n)$ is the "opposite" truth value of $f_n(n)$.

Now compare g with each f_k . At input k ,

$$g(k) \neq f_k(k)$$

by construction. Therefore, $g \neq f_k$ for every k since they differ at least at the argument k .

But g is itself a total function $N \rightarrow \{T, F\}$, so $g \in S$. This contradicts the assumption that $f_0, f_1, f_2, \dots$ lists all elements of S .

Hence S is uncountable.

$$A = C \cup C' = N$$

N^0 $\cup$
(2) Is the class of computable sets closed under the union operation? Prove your answer.

universal TM: $N(i, x) = M_i(x)$ | • computable set = Turing machine
• Each Turing Machine = One set

NO, the class of computable sets closed under the union operation. $\times$

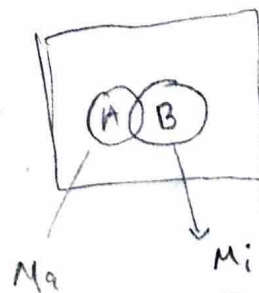
Proof: Since N^0 are countably infinite, we can show using the theorem and $A = C \cup C' = N$

Since subsets of N are recursive and all finite subsets is countable. The universal TM which is $N(i, x) = M_i(x)$ which make two tape working together. Therefore by bijection which one-one that matches two sets. This show that no class of computable set closed under union operation.

computable sets have TM recognizers run in unison. if either halt, element in union.

+10 proof attempt.

$\frac{10}{25}$



(3) Is the set $Q = \{ i \text{ such that } |M_i| \text{ accepts at least } 73i \text{ inputs} \}$ an r.e. set? Prove it.

$$Q = \{ i \text{ such that } |M_i| \text{ accepts } \geq 73i \text{ inputs} \}$$

PROOF:

For r.e we can enumerate M_i and if more than i

distinct elements appear, accept inputs so make ~~Q~~ is r.e.

For the recursive $|M_i| \geq 73i$ is a non-trivial semantic property of r.e sets. By Rice's Theorem, properties are undecidable. This equal to the complement $|M_i| \leq 73i$

is not r.e. Therefore By Rice theorem, recursive ?

If and only if both it and its complement are r.e

wrong proof for re. use snarkwalk. -10

15
25

∞ (infinite)

(4) Prove that any nonempty computable set can be enumerated.

Proof :

Recursive is solvable membership and every infinite recursive set can be enumerated.

Assume in increasing order: let $A \subseteq \mathbb{N}$ can be finite and recursive which is decidable.

$$\text{let } a(-1) = -1, \quad a(n) = \min \{x > x_{\text{old}} \mid x \in A\}$$

This decidability of A let us search for the next element and the infinite guarantees the search eventually find a 1. This gives total computability in increasing of enumeration A .

$\frac{25}{25}$